

Depth-Two Smith Profiles and Carry Geometry for Affine Hjelmslev–Radon Incidence

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Abstract

For a prime p , let B_2 be the integer point–line incidence matrix of the affine plane over $\mathbb{Z}/p^2\mathbb{Z}$, with primitive normal directions taken modulo units. The nonunit Smith factors are determined for every p :

$$\operatorname{coker}(B_2^\top)_{(p)} \cong (\mathbb{Z}/p\mathbb{Z})^{p^3(p-1)/2} \oplus (\mathbb{Z}/p^2\mathbb{Z})^{p(p-1)^2(p+2)/4} \oplus (\mathbb{Z}/p^3\mathbb{Z})^{p(p-1)/2}.$$

The canonical transfer extension is nonsplit at depth two. At arbitrary depth the relative quotient is presented by an exact two-chart cyclotomic block and a typed cross-chart class; a compatible clipped-line matrix represents its carry. The affine profile is derived here from explicitly cited cyclic and complete-ideal inputs, without transferring results from adjacent projective-rank matrices. Two such adjacent sources were available only at metadata-and-abstract level, so no theorem from them and no passage from their matrices to B_2 is used.

Keywords: Smith normal form; affine Hjelmslev geometry; Radon transform; incidence matrix; residue ring; integer lattice.

MSC 2020: 05B20 (primary); 15B36, 51E15 (secondary).

Repository: <https://github.com/aconsciousfractal/FCIG-Depth-Two-Smith-Profiles-and-Carry-Geometry-for-Affine-Hjelmslev-Radon-Incidence>

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1 Contribution and evidence boundary

The object is the integral incidence lattice of all affine fibres over the residue ring $\mathbb{Z}/p^n\mathbb{Z}$, not a matrix over a field of the same cardinality and not a projective incidence matrix. The main depth-two result is the all-prime profile in theorem 7.1. The structural result in theorem 8.3 separates the normalization problem into two cyclic diagonal blocks and one cross-chart extension class, then places the carry in compatible coordinates. The diagonal elementary divisors themselves belong to Weber–Künzer’s cyclic Wedderburn calculation [16, Propositions 3.8 and 3.15]; the residual statement here is the affine two-chart assembly, its cross block, and the geometric carry.

The proofs below establish the stated results. The finite computations listed in section 10 are bounded computational cross-checks only. The theorem includes $p = 5$. The companion does not execute the memory-intensive full-matrix check at $(p, n) = (5, 2)$;

this computational resource boundary is not a mathematical exception and is not used in the proof. No unexecuted experimental route is used. The bounded source search described in sections 3 and 11 is not an exhaustive literature search. Exact text and conventions for Landjev 2017 and Bajalan–Landjev–Rousseva 2023 were unavailable; their abstracts are used only to delimit adjacent subject matter [1, 8]. The affine integer proof below is therefore the sole proof offered here for the stated profile.

The presentation is organized as follows. Sections 2–4 fix the lattice and its maximal-order depth transition. Sections 5–7 reduce and evaluate the three depth-two invariants. Section 8 treats the relative shell, nonsplitting, two-chart block, and carry. Section 9 records the limited all-depth perspective. The final two sections give compact source, limitation, and reproducibility boundaries.

The logical spine is intentionally narrower than the exposition order. The depth-two theorem uses theorems 5.2, 6.2 and 6.3, but not the fine-kernel theorem theorem A.3. The relative-shell theorem uses the maximal-order and depth-transition layers together with the transfer intersection; it combines with theorem 7.1 only for the nonsplitting corollary. The two-chart construction in theorem 8.3 is independent of nonsplitting. Its exact depth-two filtered-rank value is separated into theorem 8.4, which additionally uses theorem 7.1; the abstract filtered identity theorem 9.1 depends only on the two-chart interface. The saturation result, theorem A.3, is therefore placed in section A as a lateral completion of the $\mathcal{K}_2$ profile.

2 Affine incidence lattice and normalization

Definition 2.1 (Affine Hjelmslev–Radon incidence). Fix a prime p and $n \geq 1$, and put

$$q = p^n, \quad \mathbf{R}_n = \mathbb{Z}/q\mathbb{Z}, \quad G_n = \mathbf{R}_n^2.$$

A vector $u = (a, b)$ is *primitive* if $a\mathbf{R}_n + b\mathbf{R}_n = \mathbf{R}_n$. Primitive vectors are identified under multiplication by $\mathbf{R}_n^\times$, and the fixed chart of directions is

$$\mathbb{P}^1(\mathbf{R}_n) = \{(1, t) : t \in \mathbf{R}_n\} \sqcup \{(ps, 1) : s \in \mathbb{Z}/p^{n-1}\mathbb{Z}\}.$$

For $u \in \mathbb{P}^1(\mathbf{R}_n)$ and $b \in \mathbf{R}_n$, let $\ell_{u,b} = \{x \in G_n : u \cdot x = b\}$. The matrix B_n has rows (u, b) , columns $x \in G_n$, and entry

$$B_n[(u, b), x] = \mathbf{1}_{\{u \cdot x = b\}}.$$

Thus B_n has $(p+1)p^{2n-1}$ rows, p^{2n} columns, and every row has p^n ones.

The chart is exhaustive because a primitive vector has a unit coordinate. When its first coordinate is a unit there is a unique representative $(1, t)$; otherwise its second coordinate is a unit and the representative is uniquely $(ps, 1)$ with s taken modulo p^{n-1} . This also proves the stated count of directions.

Let $\mathbf{A}_n = \mathbb{Z}[G_n]$, with the point basis $[x]$, and for a primitive u set

$$H_u = \ker(u : G_n \longrightarrow \mathbf{R}_n), \quad N_{H_u} = \sum_{h \in H_u} [h], \quad \mathbf{L}_n = \sum_u \mathbf{A}_n N_{H_u}.$$

All lattices in this article are additive lattices; in particular, $\mathbf{L}_n$ is never replaced by an ideal of a maximal order.

Lemma 2.2 (Group-ring dictionary). *There is a canonical isomorphism of abelian groups*

$$\text{coker}(B_n^T) \cong Q_n := A_n/L_n.$$

Proof. For a fixed u , every row of direction u is a translate of N_{H_u} . The q translates have disjoint supports and form a $\mathbb{Z}$ -basis of $A_n N_{H_u}$. Summing over the direction chart shows that the image of B_n^T in the point basis is exactly L_n . $\square$

3 Known layers and source boundaries

Proposition 3.1 (Imported transform context). *The affine transform, its normal operator, spectrum, and inversion are known in the residue-ring setting [13, Theorems 5.1, 5.3, 5.4]; the maximal-cyclic-subgroup formulation is part of the finite-group Radon framework [5, Section 7.3]. At $n = 1$, B_1 is the affine point–line incidence matrix over $\mathbb{F}_p$, whose invariant factors are covered by Chandler–Sin–Xiang [2]. These statements are context, not contributions of this paper.*

Projective Hjelmslev incidence ranks form an adjacent literature. The rational-rank result of Landjev–Vandendriessche does not give the modular or integer data of B_n [9, Theorem 5]. Kohnert’s reduction maps, neighbourhoods, and complete-fibre lifts supply broad geometric antecedents for passage between levels [6, pp. 270–272 and Lemma 2]; they do not state the reverse additive-lattice intersection proved in theorem 6.3. Broad recursive construction of projective Hjelmslev planes also predates this work [4, Theorem 2, pp. 291–294].

Proposition 3.2 (Exact modular object boundary). *After reduction modulo p , repetitions among fine affine rows do not change the row space. The repeated-row object and its generalized-polynomial bounds are treated by Laba–Trainor [7, Theorem 1.3]; Dvir records the corresponding distinct-row formulation and current bounds [3]. Neither source is used as an integer invariant-factor theorem.*

The two sources available only through metadata and abstracts report projective modular-rank questions for small chain rings and for nilpotency index two [1, 8]. Their definitions, hypotheses, and matrix conventions were not accessible. No theorem is imported from either one; no passage from those projective matrices to B_n is asserted; and their rank statements are not used to infer the integer invariant factors below. This is an explicit limitation of the present source record.

4 Maximal order and the canonical depth transition

Identify the character group of G_n with G_n by the dot pairing. A rational Galois orbit of characters of exact order p^k is the set of generators of a cyclic subgroup C of order p^k . These subgroups form a rooted projective character tree $\mathcal{T}_n$: level k has

$$d_k = (p + 1)p^{k-1}$$

nodes, each carrying the order $\mathbb{Z}[\zeta_{p^k}]$ of rank $\varphi(p^k) = (p - 1)p^{k-1}$. Let

$$\mathcal{O}_n = \mathbb{Z} \oplus \bigoplus_{k=1}^n \bigoplus_{C \in \mathcal{T}_n(k)} \mathbb{Z}[\zeta_{p^k}]$$

be the maximal order in $\mathbb{Q}[G_n]$.

Lemma 4.1 (Rational decomposition and normalization index). *The rational group algebra has one trivial component and $(p+1)p^{k-1}$ copies of $\mathbb{Q}(\zeta_{p^k})$ for each $1 \leq k \leq n$. Moreover*

$$v_p[\mathcal{O}_n : \mathbf{A}_n] = \frac{np^{2n}}{2} + \frac{(p+2)(p^{2n}-1)}{2(p^2-1)}.$$

Proof. There are $p^{2k} - p^{2k-2}$ characters of exact order p^k and every rational orbit has $\varphi(p^k)$ elements; division gives d_k . The displayed index follows by comparing the group-ring trace discriminant with the product of the cyclotomic discriminants in these components. Using $v_p(\text{disc } \mathbb{Q}(\zeta_{p^k})) = p^{k-1}(k(p-1)-1)$ for $k \geq 1$, multiplying by d_k , adding the trivial component, and simplifying the resulting geometric sums gives the formula. $\square$

For each leaf C of order p^n , let $\mathbf{R}_C$ be the image of the cyclic group ring along the root-to-leaf components, and set $\mathbf{M}_n = \sum_C \mathbf{R}_C \subset \mathcal{O}_n$. The choice of a generator of C does not affect this lattice.

Theorem 4.2 (Maximal-order quotient). *For every prime p and $n \geq 1$,*

$$\mathcal{O}_n / \mathbf{L}_n \cong \mathbb{Z}/p^{2n}\mathbb{Z} \oplus \bigoplus_{k=1}^n (\mathbb{Z}/p^{2n-k}\mathbb{Z})^{p^{2k}-p^{2k-2}}, \quad (1)$$

$$v_p[\mathcal{O}_n : \mathbf{L}_n] = np^{2n} + \frac{p^{2n}-1}{p^2-1}, \quad (2)$$

$$v_p[\mathbf{A}_n : \mathbf{L}_n] = \frac{np^{2n}}{2} - \frac{p(p^{2n}-1)}{2(p^2-1)}. \quad (3)$$

Proof. For $H = \ker(u)$ choose s with $u(s) = 1$. At a character χ ,

$$\chi(s^j N_H) = \chi(s)^j \sum_{h \in H} \chi(h).$$

The sum is p^n precisely on the cyclic annihilator $H^\perp$ and is zero elsewhere. Hence the maximal-order Fourier embedding gives the exact additive identity

$$\mathbf{L}_n = p^n \mathbf{M}_n \quad \text{inside } \mathcal{O}_n. \quad (4)$$

Split $\mathcal{O}_n = \mathcal{O}_{<n} \oplus \mathcal{O}_{(n)}$. Every top component belongs to one leaf, so $\mathbf{M}_n \rightarrow \mathcal{O}_{(n)}$ is onto. On a cyclic branch the kernel of projection to its top component is generated by $\Phi_{p^n}(t)$; at every lower root of unity,

$$\Phi_{p^n}(\zeta_{p^k}^a) = p \quad (0 \leq k < n, (a, p) = 1).$$

The top components are disjoint from leaf to leaf. Therefore, if a sum of branch elements has zero projection to $\mathcal{O}_{(n)}$, its top component vanishes on each leaf separately; cancellation between different leaves is impossible. The branch-kernel calculation may consequently be applied leafwise before the truncated branches are summed. Consequently

$$\mathbf{M}_n \cap \mathcal{O}_{<n} = p \mathbf{M}_{n-1}. \quad (5)$$

The p children of one parent induce the same truncated branch order, so no extra factor is introduced in (5). The cyclic CRT layer is prior art [12, Theorem 2.6 and Lemma 2.5]; the assembly over the projective character tree is the point requiring (5).

Induction now shows that $\mathbf{M}_n \subset \mathcal{O}_n$ has exponent n once at the root and exponent $n-k$ with multiplicity $p^{2k} - p^{2k-2}$ at level k . Scaling by p^n in (4) proves (1). Summing exponents gives (2); subtracting the index in theorem 4.1 gives (3). $\square$

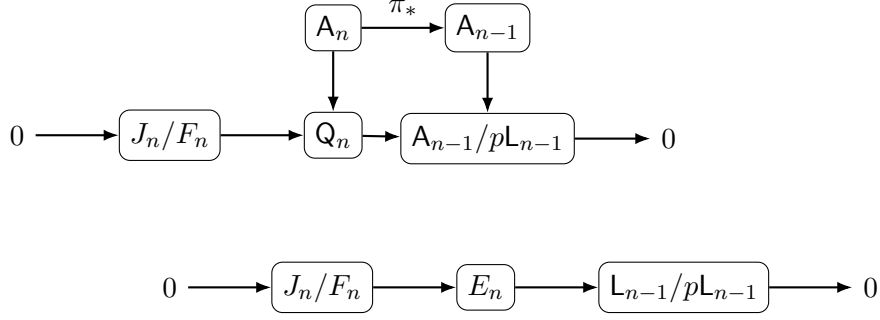


Figure 1: The canonical depth transition. The upper square records $\pi_*(\mathbf{L}_n) = p\mathbf{L}_{n-1}$ and the first quotient sequence; the lower row isolates the additional saturation layer.

Let $\pi_n : G_n \rightarrow G_{n-1}$ be reduction, let $\pi_* : \mathbf{A}_n \rightarrow \mathbf{A}_{n-1}$ be the point-basis pushforward, and put

$$\begin{aligned} J_n &= \ker \pi_*, & F_n &= J_n \cap \mathbf{L}_n, & P_n &= \pi_*^{-1}(\mathbf{L}_{n-1}), \\ \mathcal{K}_n &= J_n/F_n, & E_n &= P_n/\mathbf{L}_n = \ker(Q_n \rightarrow Q_{n-1}). \end{aligned}$$

Theorem 4.3 (Canonical depth transition). *For $n \geq 2$,*

$$\pi_*(\mathbf{L}_n) = p\mathbf{L}_{n-1},$$

and there are canonical exact sequences

$$0 \longrightarrow J_n/F_n \longrightarrow Q_n \longrightarrow \mathbf{A}_{n-1}/p\mathbf{L}_{n-1} \longrightarrow 0, \quad (6)$$

$$0 \longrightarrow J_n/F_n \longrightarrow E_n \longrightarrow \mathbf{L}_{n-1}/p\mathbf{L}_{n-1} \longrightarrow 0. \quad (7)$$

Furthermore

$$\begin{aligned} v_p|J_n/F_n| &= \frac{p^{2n-2}(n(p^2-1)-p-1)}{2}, \\ v_p|E_n| &= \frac{p^{2n-2}(n(p^2-1)-p+1)}{2}. \end{aligned}$$

Proof. For $H_u = \ker(u : G_n \rightarrow \mathbf{R}_n)$, reduction maps H_u onto $H_{\bar{u}}$ with a kernel of order p . Therefore $\pi_*(N_{H_u}) = pN_{H_{\bar{u}}}$; primitive directions lift, so these images generate exactly $p\mathbf{L}_{n-1}$. The first exact sequence is the quotient sequence for the surjection π_* , and composing it with $\mathbf{A}_{n-1}/p\mathbf{L}_{n-1} \rightarrow Q_{n-1}$ gives the second. Since $\mathbf{L}_{n-1}$ has rank p^{2n-2} , its quotient by $p\mathbf{L}_{n-1}$ is $(\mathbb{Z}/p\mathbb{Z})^{p^{2n-2}}$. The two order formulas follow by subtracting (3) at successive depths.

For completeness, choose a point-basis section $\sigma : \mathbf{A}_{n-1} \rightarrow \mathbf{A}_n$. If $c \in p\mathbf{L}_{n-1}$ and $\ell \in \mathbf{L}_n$ satisfies $\pi_*(\ell) = c$, then

$$\delta_n(c) = [\ell - \sigma(c)] \in J_n/F_n$$

is well defined and

$$Q_n \cong ((J_n/F_n) \oplus \mathbf{A}_{n-1}) / \langle (\delta_n(c), c) : c \in p\mathbf{L}_{n-1} \rangle.$$

Changing σ changes δ_n by a coboundary. No splitting is asserted. $\square$

5 Depth two and the three residual invariants

At depth two abbreviate

$$\mathcal{K}_2 = J_2/F_2, \quad E_2 = P_2/\mathbb{L}_2.$$

Let the point-fibre transfer be

$$\tau : \mathbf{A}_1 \longrightarrow \mathbf{A}_2, \quad \tau([a]) = \sum_{\pi(x)=a} [x].$$

Writing $\theta : E_2 \rightarrow \mathbb{L}_1/p\mathbb{L}_1$ for the epimorphism in (7), define

$$m_p = \text{rank}_{\mathbb{F}_p}(B_2), \quad u_p = \dim_{\mathbb{F}_p} \theta(E_2[p]), \quad w_p = \dim_{\mathbb{F}_p}(p^2\mathbf{Q}_2). \quad (8)$$

Lemma 5.1 (Finite p -group profile recovery). *Let a full-rank inclusion of free abelian groups of rank N have Smith exponents $0, 1, \dots, h$ with multiplicities $\nu_0, \nu_1, \dots, \nu_h$. If its finite cokernel is G , then*

$$\sum_{i=0}^h \nu_i = N, \quad \sum_{i=1}^h i\nu_i = v_p|G|, \quad \nu_h = \dim_{\mathbb{F}_p}(p^{h-1}G). \quad (9)$$

Consequently, an exponent- p^2 profile is fixed by N , ν_0 , and $v_p|G|$; an exponent- p^3 profile is fixed by those data together with $\dim_{\mathbb{F}_p}(p^2G)$.

Proof. In Smith coordinates, $G \cong \bigoplus_{i=1}^h (\mathbb{Z}/p^i\mathbb{Z})^{\nu_i}$. Counting coordinates, taking orders, and multiplying by p^{h-1} give the three identities in (9). For $h = 2$ the first two are a nonsingular two-equation system for ν_1, ν_2 ; for $h = 3$ the last identity first fixes ν_3 , leaving the same two-equation system. $\square$

Theorem 5.2 (Depth-two structural reduction). *For every prime p ,*

$$p^2P_2 \subset \mathbb{L}_2, \quad p^3\mathbf{A}_2 \subset \mathbb{L}_2, \quad \mathbb{L}_2 + p^2\mathbf{A}_2 = \mathbb{L}_2 + \tau(\mathbf{A}_1).$$

Consequently E_2 , $\mathcal{K}_2$, and $\mathbf{Q}_2$ have exponents at most p^2 , p^2 , and p^3 , respectively, and their complete profiles are determined by (m_p, u_p, w_p) . If unit factors are retained, the multiplicities for $\mathbb{L}_2 \subset \mathbf{A}_2$ are

$$\begin{aligned} \mu_0 &= m_p, \\ \mu_1 &= p^4 - 2m_p + \frac{p(p^2 + 1)}{2} + w_p, \\ \mu_2 &= m_p - \frac{p(p^2 + 1)}{2} - 2w_p, \quad \mu_3 = w_p. \end{aligned} \quad (10)$$

For E_2 the nonunit multiplicities are

$$\begin{aligned} d_1 &= \frac{p^2(2p^2 + p + 1)}{2} - 2m_p, \\ d_2 &= m_p - \frac{p^2(p + 1)}{2}, \end{aligned} \quad (11)$$

and for $\mathcal{K}_2$ they are

$$\begin{aligned} k_1 &= \frac{p^2(2p^2 + p + 3)}{2} - 2m_p - 2u_p, \\ k_2 &= m_p - \frac{p^2(p + 3)}{2} + u_p. \end{aligned} \quad (12)$$

Proof. For a fine point x , let S_x be the sum of all fine line rows through x and let $f_{\bar{x}} = \tau([\bar{x}])$. Counting lines through pairs of points gives the coefficientwise identity

$$S_x = \mathbf{1}_{A_2} + (p-1)f_{\bar{x}} + p^2[x]. \quad (13)$$

The all-one vector and every transferred coarse-line tube lie in L_2 ; also $p\tau(A_1) \subset L_2$ because $pA_1 \subset L_1$ at depth one. Subtracting (13) inside a point fibre gives $p^2J_2 \subset L_2$, because the point-basis differences inside the fibres generate J_2 . For a coarse line R , choose one fine point above every $\bar{x} \in R$ and write $\sigma(R)$ for their sum. Summing (13) over these p chosen points gives

$$\sum_{\bar{x} \in R} S_x = p\mathbf{1}_{A_2} + (p-1)\tau(R) + p^2\sigma(R).$$

The first two terms on the right lie in L_2 , so $p^2\sigma(R) \in L_2$. Since the coarse line rows generate L_1 , every element of $P_2 = \pi_*^{-1}(L_1)$ is a sum of an element of J_2 and such section lifts. This proves $p^2P_2 \subset L_2$.

Equation (13) also gives $p^2A_2 \subset L_2 + \tau(A_1)$. Conversely, for $f \in \tau(A_1)$ the same equation places $(p-1)f$ in $L_2 + p^2A_2$, while $pf \in L_2$; hence $f = pf - (p-1)f \in L_2 + p^2A_2$. This proves the equality of superlattices. Multiplying (13) by p and again using $p\tau(A_1) \subset L_2$ gives $p^3[x] \in L_2$ for every point x .

The equality $L_2 \cap pA_2 = L_2 \cap pP_2$ follows by applying π_* and cancelling p in the free lattice A_1 . Hence the image ranks of L_2 in A_2/pA_2 and in P_2/pP_2 coincide, so the unit Smith multiplicity for both $Q_2 = A_2/L_2$ and $E_2 = P_2/L_2$ is m_p .

We now apply theorem 5.1 rather than suppressing the profile calculation. From (3) at $n = 2$,

$$s_Q := v_p|Q_2| = p^4 - \frac{p(p^2+1)}{2}.$$

Writing the four Smith multiplicities as $\mu_0, \dots, \mu_3$, the preceding paragraph and (8) give

$$\mu_0 = m_p, \quad \mu_3 = w_p, \quad \mu_1 + \mu_2 = p^4 - m_p - w_p, \quad \mu_1 + 2\mu_2 = s_Q - 3w_p.$$

Solving the last two equations gives exactly (10).

The second exact sequence in (7) specializes to

$$0 \longrightarrow K_2 \longrightarrow E_2 \xrightarrow{\theta} (\mathbb{Z}/p\mathbb{Z})^{p^2} \longrightarrow 0. \quad (14)$$

The order formula in theorem 4.3 gives

$$s_E := v_p|E_2| = \frac{p^2(2p^2 - p - 1)}{2}.$$

If $E_2 \cong (\mathbb{Z}/p)^{d_1} \oplus (\mathbb{Z}/p^2)^{d_2}$, then

$$d_1 + d_2 = p^4 - m_p, \quad d_1 + 2d_2 = s_E.$$

Their unique solution is (11).

For completeness, put $v = p^2$ and $u = \dim_{\mathbb{F}_p} \theta(E_2[p]) = u_p$. In bases adapted first to the restriction of θ to the order- p summands and then to its quotient image, u of the d_1 order- p generators and $v - u$ of the d_2 order- p^2 generators map to a basis of $(\mathbb{Z}/p)^v$. The former leave $d_1 - u$ order- p kernel generators; each of the latter leaves its p -multiple in the kernel; and the unused order- p^2 generators remain unchanged. Thus

$$\ker \theta \cong (\mathbb{Z}/p)^{d_1+v-2u} \oplus (\mathbb{Z}/p^2)^{d_2-v+u}.$$

Substituting (11), $v = p^2$, and $u = u_p$ proves (12) and completes the profile recovery. $\square$

6 Evaluation of the depth-two invariants

The modular rank is evaluated through the annihilator of the affine line code. The reduction needed for the complete-ideal calculation is recorded explicitly first.

Lemma 6.1 (Bundle-annihilator reduction). *Put $k = \mathbb{F}_p$, $q = p^2$, and*

$$A_k = k[(\mathbb{Z}/q\mathbb{Z})^2] \cong k[x, y]/(x^q, y^q),$$

where x, y are augmentation coordinates. Let $\mathcal{I} \subset A_k$ be the modular affine-line code and let $\mathcal{J} = \text{Ann}_{A_k}(\mathcal{I})$. For each $d \in \mathbb{P}^1(k)$ there is a regular parameter pair (z_d, v_d) , with distinct tangent line $\ell_d = \text{in}(z_d) = 0$, such that

$$\mathcal{J}_d := (z_d, v_d^p)^p A_k, \quad \mathcal{J} = \bigcap_{d \in \mathbb{P}^1(k)} \mathcal{J}_d, \quad m_p = \text{length}_{A_k}(A_k/\mathcal{J}). \quad (15)$$

Proof. If $H = \langle h \rangle$ is a primitive line subgroup, then in characteristic p

$$N_H = \sum_{j=0}^{q-1} h^j = (h-1)^{q-1}, \quad \text{Ann}_{A_k}(N_H) = (h-1)A_k.$$

The translates of the N_H generate $\mathcal{I}$. The coefficient-of-identity pairing makes A_k a symmetric Frobenius algebra, so annihilating the sum of the line ideals means annihilating every one:

$$\mathcal{J} = \bigcap_H (h-1)A_k, \quad \dim_k \mathcal{I} = \dim_k(A_k/\mathcal{J}).$$

Reduction modulo p partitions the primitive directions into $p+1$ bundles, each containing p lifts. In the bundle over d , choose a group basis (g_d, t_d) for which the lifted line subgroups are generated by

$$h_{d,a} = g_d t_d^{pa} \quad (a \in k).$$

Set $z_d = g_d - 1$, $v_d = t_d - 1$, and $w_d = t_d^p - 1 = v_d^p$. The subalgebra

$$C_d = k[z_d, w_d]/(z_d^q, w_d^p)$$

has A_k free over it with basis $1, v_d, \dots, v_d^{p-1}$; hence finite ideal intersections commute with extension from C_d to A_k . In C_d write

$$f_{d,a} = (1 + z_d)(1 + w_d)^a - 1 = h_{d,a} - 1.$$

Modulo $(f_{d,a})$ one has $z_d = (1 + w_d)^{-a} - 1$ and $w_d^p = 0$, so every monomial of total (z_d, w_d) -degree at least p vanishes. Thus $(z_d, w_d)^p \subseteq \bigcap_a (f_{d,a})$.

For the reverse inclusion, pass to $C_d/(z_d, w_d)^p$. The truncated series

$$Z_d = \log(1 + z_d), \quad W_d = \log(1 + w_d)$$

use only the invertible denominators $1, \dots, p-1$ and define a coordinate automorphism. Under it, $(f_{d,a})$ becomes the linear ideal $(Z_d + aW_d)$. A homogeneous polynomial of degree below p lying in all these p distinct linear ideals is divisible by $\prod_{a \in k} (Z_d + aW_d)$, of degree p , and is therefore zero. Consequently

$$\bigcap_{a \in k} (h_{d,a} - 1)A_k = (z_d, v_d^p)^p A_k.$$

Intersecting these equalities over the $p+1$ coarse directions proves the first part of (15); the Frobenius dimension identity proves the second. $\square$

Theorem 6.2 (Exact modular rank and thick-kernel profile). *For every prime p ,*

$$m_p = \left(\frac{p(p+1)}{2} \right)^2.$$

Consequently

$$E_2 \cong (\mathbb{Z}/p\mathbb{Z})^{p^3(p-1)/2} \oplus (\mathbb{Z}/p^2\mathbb{Z})^{p^2(p^2-1)/4}.$$

Proof. The exact modular object and its known upper-bound setting are those of Łaba–Trainor and Dvir [3, 7]; equality is established here by the following complete-ideal calculation. By theorem 6.1, it is enough to compute the colength of the displayed annihilator. Lift it to the complete regular local ring $R = k[[x, y]]$, with maximal ideal $\mathfrak{m} = (x, y)$, and set

$$I_d = (z_d, v_d^p)^p, \quad I = \bigcap_d I_d.$$

A monomial outside $(z, v^p)^p$ satisfies $a + \lfloor b/p \rfloor \leq p - 1$, hence $a + b \leq p^2 - 1$. Thus $\mathfrak{m}^{p^2} \subset I$, the Artinian truncation introduces no extra ideal, and

$$\text{length}_{A_k}(A_k/\mathcal{J}) = \text{length}_R(R/I). \quad (16)$$

The Newton inequality for I_d is $pa + b \geq p^2$, so every I_d is integrally closed; their finite intersection I is integrally closed as well. The complete-ideal/antinef result cited below is stated over an algebraically closed residue field. Let $\bar{k}$ be an algebraic closure of k . Because all the ideals contain $\mathfrak{m}^{p^2}$, scalar extension may be performed in the finite-dimensional quotient $R/\mathfrak{m}^{p^2}$. Faithful flatness makes it commute with the finite intersection, and

$$\dim_k(R/I) = \dim_{\bar{k}}(\bar{k}[[x, y]]/I\bar{k}[[x, y]]).$$

We may therefore apply the geometric theorem after base change and descend the resulting integer colength to k .

The resolution of each $(z, v^p)^p$ has multiplicity p at the origin and at a free chain of $p - 1$ infinitely near points on the tangent $z = 0$. The $p + 1$ tangent lines are distinct, so the chains separate after the first blow-up. In the strict exceptional basis, the divisor of one I_d has coefficient p at the root, coefficient $p(j + 1)$ on its own level- j arm point, and coefficient p on the other arms. Intersecting the global-section ideals takes the componentwise maximum. Thus, on the simultaneous star-shaped resolution, the divisor whose global-section ideal is I has root coefficient p and coefficient $p(j + 1)$ at level j on every arm.

The complete-ideal/antinef correspondence and the Hoskin–Deligne colength formula are used in the precise form developed by Lipman and Lipman–Watanabe [10, Section 1][11, Corollary 1.1.2 and Lemma 1.2]. For odd p , put

$$h = \frac{p-1}{2}, \quad A_0 = \frac{p^2 + 2p - 1}{2}.$$

The least antinef majorant has strict exceptional coefficients

$$c_0 = A_0, \quad c_{d,j} = A_0 + jh \quad (1 \leq j \leq p-1),$$

and therefore point basis A_0 at the root and h at every arm point. To see minimality, let $b_0, b_{d,j}$ be any antinef majorant and write $u_{d,j} = b_{d,j} - b_{d,j-1}$. Then

$$u_{d,1} \geq \cdots \geq u_{d,p-1} \geq 0, \quad b_0 \geq \sum_d u_{d,1}, \quad b_{d,p-1} \geq p^2.$$

If $b_0 < A_0$, the endpoint bound forces every $u_{d,1} \geq h + 1$, in contradiction with the root inequality. Discrete concavity along each arm then places every $b_{d,j}$ above the chord $A_0 + jh$. This proves minimality without a symmetry assumption.

When $p = 2$, the unresolved divisor has strict coefficients 2 at the root and 4 at each of the three endpoints. If b_0 and $b_{d,1}$ are the coefficients of an antinef majorant, then $b_0 \geq \sum_d (b_{d,1} - b_0)$, so $b_0 \geq 3$. Root coefficient 3 and endpoint coefficients 4 attain the bound; their point basis is 3 at the root and 1 at each arm point. Hoskin–Deligne now gives

$$\text{length}(R/I) = \begin{cases} \binom{4}{2} + 3\binom{2}{2} = 9, & p = 2, \\ \binom{A_0 + 1}{2} + (p + 1)(p - 1)\binom{h + 1}{2} = \frac{p^2(p + 1)^2}{4}, & p \text{ odd.} \end{cases}$$

Together with (16), this proves the rank formula. Substitution in (11) proves the displayed profile.

We record why the same calculation also proves the initial-ideal equality used later. Put $S_0 = k[X, Y]$ and, for the initial regular pair (ℓ_d, m_d) , set

$$I_d^{\text{lin}} = (\ell_d, m_d^p)^p, \quad I^{\text{lin}} = \bigcap_d I_d^{\text{lin}}.$$

The ideals I_d and I_d^{lin} have the same weighted proximity graph: multiplicity p at the origin followed by a free chain of $p - 1$ points on tangent d . Their intersections are complete ideals obtained by the componentwise maximum of the branch divisors. Although the actual and linearized infinitely near points need not coincide, their common star-shaped proximity graph and divisor coefficients are exactly the data used by unloading and by Hoskin–Deligne. The least antinef majorant computed above therefore gives

$$\text{length}_R(R/I) = \text{length}_{S_0}(S_0/I^{\text{lin}}) = \frac{p^2(p + 1)^2}{4}. \quad (17)$$

For the maximal-ideal filtration, finite colength gives

$$\text{length}_R(R/I) = \text{length}_{S_0}(S_0/\text{in } I).$$

On the other hand, initial forms always give the inclusion

$$\text{in } I \subseteq I^{\text{lin}} = \bigcap_d \text{in } I_d.$$

The two quotients have equal finite length by (17); hence the inclusion is equality. Since all ideals contain the relevant p^2 -nd augmentation power, passage back to the Artinian group-algebra quotient gives

$$\text{in} \left(\bigcap_d \mathcal{J}_d \right) = \bigcap_d \text{in}(\mathcal{J}_d). \quad (18)$$

Thus no general commutation rule for initial ideals and intersections is being assumed. $\square$

The remaining invariant u_p controls only the fine kernel $\mathcal{K}_2$, not the main quotient $\mathbf{Q}_2$. Its complete Bockstein and filtered-interpolation proof is placed on the lateral branch in section A; it is not an input to theorem 7.1.

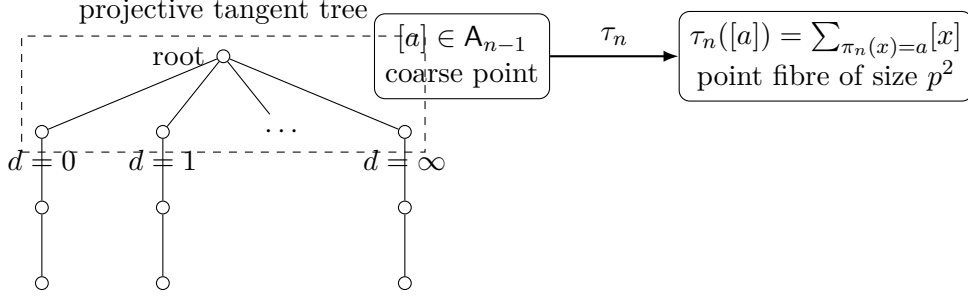


Figure 2: The $p+1$ separated tangent arms in the complete-ideal calculation (left) and the point-fibre transfer used in the reverse lattice intersection (right). The two mechanisms evaluate m_p and w_p , respectively.

Theorem 6.3 (Exact point-fibre transfer). *For every prime p and $n \geq 2$,*

$$\tau_n(\mathbf{A}_{n-1}) \cap \mathbf{L}_n = \tau_n(\mathbf{L}_{n-1}).$$

Consequently τ_n induces an injection $\mathbf{Q}_{n-1} \hookrightarrow \mathbf{Q}_n$, and at depth two

$$w_p = \dim_{\mathbb{F}_p}(p^2\mathbf{Q}_2) = \frac{p(p-1)}{2}.$$

Proof. Let $\mathcal{F}_n : \mathbf{A}_n \rightarrow \mathcal{O}_n$ be the maximal-order Fourier embedding and write $\mathcal{O}_n = \mathcal{O}_{<n} \oplus \mathcal{O}_{(n)}$. A point fibre has p^2 points. Character summation on its kernel gives

$$\mathcal{F}_n(\tau_n(a)) = p^2 \iota_n(\mathcal{F}_{n-1}(a)) \in \mathcal{O}_{<n}; \quad (19)$$

the exact-order- p^n component vanishes. On the other hand, theorem 4.2 gives

$$\mathbf{L}_n \cap \mathcal{O}_{<n} = p^n(\mathbf{M}_n \cap \mathcal{O}_{<n}) = p^{n+1}\mathbf{M}_{n-1}.$$

Since the ambient orders are torsion-free, cancellation in (19) yields

$$\tau_n(a) \in \mathbf{L}_n \iff \mathcal{F}_{n-1}(a) \in p^{n-1}\mathbf{M}_{n-1} \iff a \in \mathbf{L}_{n-1}.$$

Distinct point fibres have disjoint support, so τ_n is injective. At depth two, theorem 5.2 identifies $p^2\mathbf{Q}_2$ with this transferred copy of $\mathbf{Q}_1$; the depth-one profile $\mathbf{Q}_1 \cong (\mathbb{Z}/p\mathbb{Z})^{p(p-1)/2}$ finishes the proof. $\square$

7 The all-prime depth-two Smith profile

Theorem 7.1 (Depth-two Smith profile). *For every prime p , the inclusion $\mathbf{L}_2 \subset \mathbf{A}_2$ has unit multiplicity*

$$\mu_0 = \frac{p^2(p+1)^2}{4},$$

and

$$\mathbf{Q}_2 \cong (\mathbb{Z}/p\mathbb{Z})^{p^3(p-1)/2} \oplus (\mathbb{Z}/p^2\mathbb{Z})^{p(p-1)^2(p+2)/4} \oplus (\mathbb{Z}/p^3\mathbb{Z})^{p(p-1)/2}. \quad (20)$$

Equivalently, these are the nonunit Smith factors of B_2 .

Table 1: All-prime Smith multiplicities for $L_2 \subset A_2$.

Smith factor	multiplicity	role in Q_2
1	$p^2(p+1)^2/4$	unit; omitted from the quotient
p	$p^3(p-1)/2$	exponent-one summands
p^2	$p(p-1)^2(p+2)/4$	exponent-two summands
p^3	$p(p-1)/2$	transferred depth-one carry

Table 2: Exact $p = 2, 3$ controls. These rows are bounded computational cross-checks, not the proof for all primes.

p	unit	p -factor	p^2 -factor	p^3 -factor
2	9	4	2	1
3	36	27	15	3

Proof. By theorems 6.2 and 6.3,

$$m_p = \frac{p^2(p+1)^2}{4}, \quad w_p = \frac{p(p-1)}{2}.$$

Substitution of m_p and w_p into (10) gives the four multiplicities in table 1; deleting the unit factors gives (20). Their weighted exponent sum equals $p^4 - p(p^2 + 1)/2$, independently matching (3) at $n = 2$. $\square$

The two rows of table 2 are reconstructed by the exact local validators described in section 10. They are deliberately presented after the uniform proof and are not used to interpolate a law in p .

8 Relative top shell, nonsplitting, and carry

Define the relative quotient and the exact top-shell projection by

$$\text{Rel}_n = Q_n / \tau_n(Q_{n-1}), \quad \lambda_n = \text{pr}_{(n)} \mathcal{F}_n, \quad \mathcal{C}_n = \lambda_n(A_n) \subset \mathcal{O}_{(n)}.$$

Theorem 8.1 (Relative top shell). *For every prime p and $n \geq 2$,*

$$\ker \lambda_n = \tau_n(A_{n-1}), \quad \lambda_n(L_n) = p^n \mathcal{O}_{(n)}, \quad (21)$$

$$\text{Rel}_n \cong \mathcal{C}_n / p^n \mathcal{O}_{(n)}, \quad p^n \text{Rel}_n = 0, \quad (22)$$

and

$$v_p | \text{Rel}_n | = \frac{p^{2n-2}(n(p^2 - 1) - p + 1)}{2}.$$

If a $\mathbb{Z}$ -linear section of λ_n is chosen, there is a homomorphism

$$\kappa_n : p^n \mathcal{O}_{(n)} \longrightarrow Q_{n-1}$$

such that

$$Q_n \cong (Q_{n-1} \oplus \mathcal{C}_n) / \langle (\kappa_n(c), c) : c \in p^n \mathcal{O}_{(n)} \rangle. \quad (23)$$

Its class is independent of the section.

Proof. Equation (19) gives the inclusion of the transfer lattice in $\ker \lambda_n$. Rationally both have dimension p^{2n-2} . The transfer lattice is primitive in $\mathbf{A}_n$ because the all-one vector in each point fibre extends to an integral basis; the kernel of a map to the free lattice $\mathcal{O}_{(n)}$ is also saturated. Hence they agree. The image identity follows from $\mathcal{F}_n(\mathbf{L}_n) = p^n \mathbf{M}_n$ and the surjectivity $\mathbf{M}_n \rightarrow \mathcal{O}_{(n)}$. This proves (21)–(22).

To derive rather than assume the top-shell normalization index, write the index from theorem 4.1 as

$$d_{p,m} := v_p[\mathcal{O}_m : \mathbf{A}_m] = \frac{mp^{2m}}{2} + \frac{(p+2)(p^{2m}-1)}{2(p^2-1)}.$$

Inside $\mathcal{O}_n = \mathcal{O}_{<n} \oplus \mathcal{O}_{(n)}$, the lower-tree intersection of $\mathcal{F}_n(\mathbf{A}_n)$ is, by the kernel equality above and (19),

$$\mathcal{F}_n(\mathbf{A}_n) \cap \mathcal{O}_{<n} = p^2 \iota_n(\mathcal{F}_{n-1}(\mathbf{A}_{n-1})).$$

The index of a full-rank lattice in a direct sum is the product of the index of its projection and the index of its intersection with the kernel. Consequently

$$\begin{aligned} v_p[\mathcal{O}_{(n)} : \mathcal{C}_n] &= d_{p,n} - d_{p,n-1} - 2p^{2n-2} \\ &= \frac{p^{2n-2}(n(p^2-1) + p - 1)}{2}. \end{aligned}$$

Subtracting this from $n \operatorname{rank} \mathcal{O}_{(n)}$, where $\operatorname{rank} \mathcal{O}_{(n)} = p^{2n} - p^{2n-2}$, gives the displayed order of Rel_n . Equivalently, the same exponent is $v_p|\mathbf{Q}_n| - v_p|\mathbf{Q}_{n-1}|$ by theorems 4.2 and 4.3. Finally, for $c \in p^n \mathcal{O}_{(n)}$ choose $\ell \in \mathbf{L}_n$ above c and a section lift $\sigma_n(c)$. The unique a satisfying $\ell - \sigma_n(c) = \tau_n(a)$ defines $\kappa_n(c) = [a]$. The transfer intersection in theorem 6.3 makes this well defined; the resulting graph is exactly (23). A section change adds the usual coboundary. $\square$

Corollary 8.2 (Nonsplitting). *For every prime p , the exact sequence*

$$0 \longrightarrow \mathbf{Q}_1 \longrightarrow \mathbf{Q}_2 \longrightarrow \operatorname{Rel}_2 \longrightarrow 0$$

does not split. In the exact finite cross-check $(p, n) = (2, 3)$,

$$\begin{aligned} \operatorname{Rel}_3 &\cong (\mathbb{Z}/2)^{13} \oplus (\mathbb{Z}/4)^{12} \oplus (\mathbb{Z}/8)^9, \\ E_3 &\cong (\mathbb{Z}/2)^6 \oplus (\mathbb{Z}/4)^{26} \oplus (\mathbb{Z}/8)^2, \end{aligned}$$

so equal order does not imply isomorphism.

Proof. At depth two, $\mathbf{Q}_1$ has exponent p and Rel_2 has exponent at most p^2 , whereas theorem 7.1 gives $p(p-1)/2 > 0$ factors of order p^3 in $\mathbf{Q}_2$. A split extension cannot have such a factor. For the second statement, the exact top-shell and fibre-coordinate profiles are, respectively,

$$0:9, 1:12, 2:13, 3:14, \quad 0:30, 1:6, 2:26, 3:2.$$

Reversing the top-shell profile as in (22) and omitting unit coordinates gives the displayed groups. Both have order 2^{64} , but their images after multiplication by 4 have dimensions 9 and 2. This finite row is a counterexample to a universal identification, not an all-depth formula. $\square$

Put $q^\flat = p^{n-1}$ and $e = \varphi(q) = q - q^\flat$. In the cyclic polynomial order $\mathcal{R}_q^X = \mathbb{Z}[X]/(X^q - 1)$, let

$$\mathcal{U}_q^X = \langle 1, X, \dots, X^{e-1} \rangle_{\mathbb{Z}}, \quad \mathfrak{K}_q^X = \Phi_q \mathcal{R}_q^X.$$

The monic factorization $X^q - 1 = \Phi_q(X)(X^{q^\flat} - 1)$ gives the unimodular decomposition $\mathcal{R}_q^X = \mathcal{U}_q^X \oplus \mathfrak{K}_q^X$. Since $\tau_n(\mathbf{A}_{n-1}) = \mathfrak{K}_q^X \otimes \mathfrak{K}_q^Y$, an integral basis of the quotient is

$$\mathbf{A}_n / \tau_n(\mathbf{A}_{n-1}) \cong (\mathcal{U}_q^X \otimes \mathcal{R}_q^Y) \oplus (\mathfrak{K}_q^X \otimes \mathcal{U}_q^Y). \quad (24)$$

Here are the integral types of the two summands and their evaluation spaces. Let $\mathcal{O}_q = \mathbb{Z}[\zeta_q]$ in the ordered power basis $1, \zeta_q, \dots, \zeta_q^{e-1}$, and put

$$D_1 = \mathcal{U}_q^X \otimes \mathcal{R}_q^Y, \quad W_1 = \mathcal{O}_q^{\oplus q}, \quad D_2 = \mathfrak{K}_q^X \otimes \mathcal{U}_q^Y, \quad W_2 = \mathcal{O}_q^{\oplus q^\flat}.$$

All four are free abelian groups, with

$$\text{rank } D_1 = \text{rank } W_1 = eq, \quad \text{rank } D_2 = \text{rank } W_2 = eq^\flat.$$

The first-chart evaluation is an injective square $\mathbb{Z}$ -linear map $V_q^{(n)} : D_1 \rightarrow W_1$; in $q \times q$ block notation its (t, j) entry is multiplication by ζ_q^{tj} on the common rank- e power basis. The second diagonal evaluation is $pV_{q^\flat}^{(n)} : D_2 \rightarrow W_2$, with (s, r) block multiplication by $p\zeta_q^{psr}$. Finally $X_{p,n} : D_1 \rightarrow W_2$ is the cross-chart evaluation. Thus every block below denotes a homomorphism between specified finite free $\mathbb{Z}$ -modules, not an untyped scalar array.

Theorem 8.3 (Two-chart block and clipped-line carry). *In integral bases compatible with (24), the inclusion $\mathcal{C}_n \subset \mathcal{O}_{(n)}$ has the exact block presentation*

$$T_{p,n} \sim \begin{pmatrix} V_q^{(n)} & 0 \\ X_{p,n} & pV_{q^\flat}^{(n)} \end{pmatrix}, \quad (25)$$

where the displayed maps are expanded in the fixed integral bases just specified. The stable cross-chart datum is

$$\omega_{p,n} = [X_{p,n}] \in \text{Ext}_{\mathbb{Z}}^1(\text{coker } V_q^{(n)}, \text{coker}(pV_{q^\flat}^{(n)})).$$

There are compatible relation and carry matrices $\mathbf{Z}_{p,n}$ and $\mathbf{C}_{p,n}^{\text{car}}$ satisfying

$$T_{p,n} \mathbf{Z}_{p,n} = \mathbf{Z}_{p,n} T_{p,n} = p^n I, \quad (26)$$

with

$$(\mathbf{C}_{p,n}^{\text{car}})_{(r,s);(u,j)} = \mathbf{1}_{\{u \cdot (e+r, e+s) = j \bmod q\}}, \quad (27)$$

where $(r, s) \in (\mathbb{Z}/q^\flat\mathbb{Z})^2$, u is a chart direction, and $0 \leq j < e$. If $U\mathbf{Z}_{p,n}V = \text{diag}(p^{a_i})$, the extension is represented by the columns $(\mathbf{C}_{p,n}^{\text{car}}V)_i$ modulo $p^{a_i}\mathbf{Q}_{n-1}$.

Proof. For a basis monomial $X^i Y^j$ in $\mathcal{U}_q^X \otimes \mathcal{R}_q^Y$, evaluation at the two direction charts gives

$$(1, t) : \zeta_q^{i+tj}, \quad (ps, 1) : \zeta_q^{psi+j}.$$

For $\Phi_q(X)X^rY^i$ in $\mathfrak{K}_q^X \otimes \mathcal{W}_q^Y$, the corresponding values are 0 and $p\zeta_q^{psr+i}$, since $\Phi_q(\zeta_q) = 0$ and $\Phi_q(\zeta_q^{ps}) = p$. Stacking these rows proves (25) by unimodular basis changes.

The exact elementary divisors of both diagonal cyclic blocks follow from Weber–Künzer [16, Proposition 3.15]; Proposition 3.8 supplies a diagonalization. Explicitly, if $j = \sum a_k p^k$, their valuation

$$\beta_p(j) = \sum_{k \geq 0} (a_k - a_{k+1})(k+1)p^k$$

equals $\sum_{d=1}^j p^{v_p(d)}$ by digit telescoping. Total ramification multiplies the level- $(n-1)$ valuation by p , and the scalar p adds $\varphi(q)$. This identifies the cited result with both diagonal blocks; it does not remove $X_{p,n}$.

Set $A = V_q^{(n)} : D_1 \rightarrow W_1$, $B = pV_{q^b}^{(n)} : D_2 \rightarrow W_2$, and $X = X_{p,n} : D_1 \rightarrow W_2$. Both A and B are injective: after tensoring with $\mathbb{Q}$ they are ordinary Fourier evaluation isomorphisms, and their domains are torsion free. Projection to W_1 therefore induces the exact sequence of finite abelian groups

$$0 \longrightarrow \text{coker } B \xrightarrow{[w_2] \mapsto [(0, w_2)]} \text{coker } T_{p,n} \xrightarrow{[(w_1, w_2)] \mapsto [w_1]} \text{coker } A \longrightarrow 0. \quad (28)$$

Indeed, if $[w_1] = 0$, write $w_1 = Ad_1$ and subtract $T_{p,n}(d_1, 0)$, leaving a class represented by $(0, w_2 - Xd_1)$; injectivity at the left follows because $Ad_1 = 0$ forces $d_1 = 0$. This is also the exact sequence obtained from the two length-one free resolutions by the snake lemma.

A domain shear $D_1 \rightarrow D_2$ and a codomain shear $W_1 \rightarrow W_2$ replace the cross map by

$$X \longmapsto X + BH + GA, \quad H \in \text{Hom}_{\mathbb{Z}}(D_1, D_2), \quad G \in \text{Hom}_{\mathbb{Z}}(W_1, W_2).$$

Conversely, these are precisely the changes of representative obtained from such integral splittings. Since A and B are injective presentations by finite free $\mathbb{Z}$ -modules, the standard presentation quotient is

$$\frac{\text{Hom}_{\mathbb{Z}}(D_1, W_2)}{B \text{Hom}_{\mathbb{Z}}(D_1, D_2) + \text{Hom}_{\mathbb{Z}}(W_1, W_2)A} \cong \text{Ext}_{\mathbb{Z}}^1(\text{coker } A, \text{coker } B).$$

Thus $[X_{p,n}]$ is exactly the extension class of (28), with its category and integral bases fixed.

Finally extend (24) by $\mathfrak{K}_q^X \otimes \mathfrak{K}_q^Y$ to a unimodular basis of $\mathbf{A}_n$. Choose, for each (u, j) , the affine-line indicator $W_{u,j}$ with $0 \leq j < e$. To make the coordinate passage explicit, write $L_{u,j}(x, y) = \mathbf{1}_{\{u \cdot (x, y) = j \bmod q\}}$ and $\widehat{t} = e + (t \bmod q^b)$. Pointwise on $(\mathbb{Z}/q\mathbb{Z})^2$ one has the telescoping identity

$$\begin{aligned} L_{u,j}(x, y) &= \mathbf{1}_{\{x < e\}} (L_{u,j}(x, y) - L_{u,j}(\widehat{x}, y)) \\ &\quad + \mathbf{1}_{\{y < e\}} (L_{u,j}(\widehat{x}, y) - L_{u,j}(\widehat{x}, \widehat{y})) + L_{u,j}(\widehat{x}, \widehat{y}). \end{aligned}$$

The first two terms are precisely the D_1 - and D_2 -coordinates of $\mathbf{Z}_{p,n}$, while the last term is the high–high coordinate (27). The top Fourier image of $W_{u,j}$ is $q\zeta_q^j$ in the u -component and zero elsewhere, so stacking these coordinate columns gives $T_{p,n}\mathbf{Z}_{p,n} = qI$. Both matrices are square; over $\mathbb{Q}$ this says $\mathbf{Z}_{p,n} = qT_{p,n}^{-1}$ and therefore $\mathbf{Z}_{p,n}T_{p,n} = qI$ as an integral equality, proving (26). A Smith change of relation basis sends $\mathbf{C}_{p,n}^{\text{car}}$ to $\mathbf{C}_{p,n}^{\text{car}}V$, and the relation p^{a_i} makes its i th column well defined modulo $p^{a_i}\mathbf{Q}_{n-1}$. $\square$

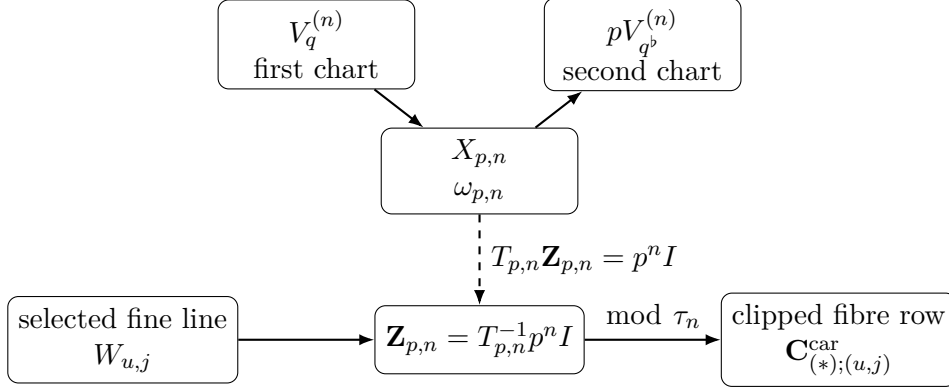


Figure 3: The two cyclic charts are glued by $\omega_{p,n}$; in the same integral basis the relation inverse transports a selected fine line to the clipped-line carry.

Corollary 8.4 (Depth-two first filtered rank). *For the two-chart class at depth two,*

$$d_0(2, 2) = 0, \quad d_0(p, 2) = \frac{(p-1)^2(p+1)}{8} \quad (p \text{ odd}).$$

Proof. The first filtered rank of the cross-chart class in theorem 8.3 is

$$d_0(p, n) = \text{rank}_{\mathbb{F}_p} \left(\frac{\overline{V_q^{(n)}}}{X_{p,n}} \right) - \text{rank}_{\mathbb{F}_p} \overline{V_q^{(n)}}.$$

At $n = 2$, the diagonal unit count is determined by the two Weber–Künzer blocks in theorem 8.3; subtracting it from the total unit multiplicity μ_0 in theorem 7.1 gives the displayed values. Thus the corollary, unlike the two-chart construction itself, depends on the depth-two Smith theorem. $\square$

9 Filtered all-depth perspective and its limits

The two-chart class admits a standard filtered description. Work over $\mathbb{Z}_p$ and let $A : D_A \rightarrow V_A$ and $B : D_B \rightarrow V_B$ be injective square lattice maps. For $r \geq 0$, put

$$M_r(A) = \{x \in D_A : Ax \in p^r V_A\}, \quad \Lambda_r(A) = \log_p[D_A : M_r(A)].$$

For $H = \begin{pmatrix} A & 0 \\ X & B \end{pmatrix}$ define

$$\Omega_r : M_r(A) \longrightarrow V_B / (BD_B + p^r V_B), \quad x \longmapsto [Xx],$$

and $\varepsilon_r(H) = \log_p |\text{im } \Omega_r|$, with $\varepsilon_0 = 0$.

Proposition 9.1 (Filtered obstruction identity and companion boundary). *For every $r \geq 1$,*

$$\Lambda_r(H) = \Lambda_r(A) + \Lambda_r(B) + \varepsilon_r(H). \quad (29)$$

If the Smith exponents of an injective map F lie in $[0, R]$, their multiplicities are recovered from

$$\begin{aligned} m_0(F) &= \Lambda_1(F), \\ m_h(F) &= \Lambda_{h+1}(F) - 2\Lambda_h(F) + \Lambda_{h-1}(F) \quad (1 \leq h < R), \\ m_R(F) &= \text{rank}(F) - \Lambda_R(F) + \Lambda_{R-1}(F). \end{aligned}$$

Applied to (25), the finite list $\varepsilon_r(T_{p,n})$, $1 \leq r \leq n$, determines the top-shell profile. Together with the carry block, the same identity yields an exact full-size recursive presentation. This last construction is a methods/software companion: it is not an independent geometry constructor, a reduced-rank algorithm, or a headline theorem of this manuscript.

Proof. A pair (x, y) lies in $M_r(H)$ exactly when

$$Ax \in p^r V_A, \quad Xx + By \in p^r V_B.$$

The first condition places x in $M_r(A)$; the second has a solution if and only if $x \in \ker \Omega_r$, and two solutions differ by $M_r(B)$. Hence

$$0 \longrightarrow M_r(B) \longrightarrow M_r(H) \longrightarrow \ker \Omega_r \longrightarrow 0.$$

Taking indices proves (29). In Smith coordinates, $\Lambda_r(F) = \sum_i \max(r - a_i, 0)$, and the displayed finite differences recover the multiplicities.

The filtration and multiplicity-recovery layer is stated directly by Sin [14, Section 2.1, equations (4)–(8)]. Moreover, Ω_r is the coordinate form of the ordinary connecting morphism

$$\operatorname{coker}(A)[p^r] \longrightarrow \operatorname{coker}(B)/p^r \operatorname{coker}(B)$$

in the long exact Tor sequence [15, Lemma 10.75.2]. Thus the abstract mechanism receives no originality credit here. The domain-specific companion computes its image lattices on the fixed affine incidence geometry and transports integral bases through

$$\mathcal{B}_n = \begin{pmatrix} \mathcal{B}_{n-1} & \Xi_n \\ 0 & \mathbf{Z}_{p,n} \end{pmatrix},$$

where $\mathcal{B}_n$ is the typed basis-transport matrix and Ξ_n its carry-coupling block; neither symbol denotes the preimage lattice P_n or the cyclotomic ideal $\mathfrak{R}_q^X$. That realization runs at ambient rank p^{2n} and relies on the separately fixed two-chart geometry. Broad recursion through Hjelmslev levels is also known geometrically [4, 6]. No reduction of asymptotic work and no uniform explicit all-depth profile is asserted. $\square$

10 Sources, limitations, and computational role

The proof of the exact affine integer statement is given in theorem 7.1. The sources in table 3 delimit the imported object, cyclic, local-algebra, and extension layers; “proved here” never means a literature-wide firstness claim.

The source search was bounded and is not a completeness certificate. Exact text and conventions for [1, 8] were unavailable; no projective matrix is identified with B_2 , no modular rank is promoted to an integer Smith profile, and no originality or priority conclusion is drawn. The software reproduces closed exact controls, rejects typed and lattice mutations, and evaluates the shipped full-size companion constructions. It is bounded computational cross-checking, not a substitute for any all-parameter proof. The uniform theorem includes $p = 5$, while the memory-intensive full-matrix companion check at $(p, n) = (5, 2)$ is deliberately not executed or reported. No unexecuted generic-orbit experiment enters this manuscript.

Table 3: Compact source and nontransfer boundary.

Layer	Sources	Boundary used here
Affine transform and depth one	[2, 5, 13]	context and field-depth factors; no new ownership
Maximal-order branch CRT	[12]	cyclic layer imported; tree assembly proved in theorem 4.2
Modular rank and complete ideals	[3, 7, 10, 11]	exact object/bounds and local-algebra machinery attributed; application proved in theorem 6.2
Level geometry	[4, 6, 9]	broad projective rank and lift geometry only; no integral affine transfer
Cyclic diagonal blocks	[16]	both diagonal Smith formulas are prior art; the cross-chart class and carry remain
Filtered extension layer	[14, 15]	standard-derived identity; domain-specific realization is supporting only
Unavailable exact text	[1, 8]	metadata/abstract scope only; no theorem or matrix passage is imported

11 Reproducibility and companion interface

The theorem proofs are contained in the manuscript. The repository companion has the narrower roles in table 4. It is self-contained, uses exact integer arithmetic through SymPy, makes no network requests, and writes no scientific result files.

From the repository root, the exact arithmetic receipt is printed by

```
python -B companion/affine_hjelmslev_radon_smith.py
```

and the hostile and differential controls are run by

```
python -B -m pytest -q -p no:cacheprovider \
  --basetemp /tmp/fcig-hjelmslev-pytest tests
```

The two package-integrity layers are checked with

```
python -B scripts/check_manifest.py
python -B scripts/check_release.py
```

The paper is built from `paper/main.tex` with LuaLaTeX, Biber, and `latexmk`; `REPROD UCE.md` records the fixed-epoch command and the precise environment boundary. The repository ships the title-named PDF. Its build and the finite computations provide reproducibility evidence only. The PDF catalog declares English and the public source preserves semantic section, theorem, equation and table structure. PDF/UA conformance is not claimed; the exact accessibility boundary is recorded in `ACCESSIBILITY.md`.

Table 4: Compact companion interface.

Layer	Reproduced object	Evidentiary boundary
Exact controls	closed Smith profiles and index identities	bounded arithmetic cross-checks only; no finite extrapolation
Hostile tests	prime types, lattice equalities, and fail-closed boundaries	rejects invalid primes, strict-sublattice mutations, and the reserved full-matrix target
Full-size companion	filtered image lattices and carry transport	exact supporting realization on fixed geometries; no finite extrapolation or compression claim
Build checks	source manifest, release digest, and clean PDF	integrity and visual evidence, not mathematical proof

A Saturation and augmentation: the lateral fine-kernel branch

Let C_i be the free abelian group on depth- i affine lines and let $\partial_i : C_i \rightarrow A_i$ be incidence. Reduction of lines gives $\gamma : C_2 \rightarrow C_1$ and

$$\pi_* \partial_2 = p \partial_1 \gamma. \quad (30)$$

Lemma A.1 (Bockstein image and dual criterion). *The image $H_p := \theta(E_2[p]) \subset L_1/pL_1$ is exactly the image of*

$$\ker(\partial_2 \bmod p) \longrightarrow L_1/pL_1, \quad c \longmapsto \partial_1 \gamma(c) \bmod pL_1. \quad (31)$$

If $Y = (L_1/pL_1)^*$, ρ reduces fine lines, and $\text{Col}_2 = \text{im}(\partial_2^\top \bmod p)$ is the fine column code, then

$$y \text{ annihilates } H_p \iff \rho^* y \in \text{Col}_2. \quad (32)$$

Proof. If $c \bmod p \in \ker(\partial_2 \bmod p)$, write $\partial_2 c = px$. Cancellation in the free lattice in (30) gives $\pi_* x = \partial_1 \gamma(c)$. Conversely, every class in $E_2[p]$ has a representative x with $px \in L_2$, so choosing c with $\partial_2 c = px$ recovers it in this way. This proves (31).

A line-coordinate vector represents an element of Y precisely when its sum on a parallel class is independent of the direction. Indeed, the differences of the $p+1$ complete parallel-class sums give p integral relations, and the coarse point Gram matrix $pI + J$ is nonsingular over $\mathbb{Q}$, so these have the full relation rank. Pairing (31) with y gives

$$\langle y, \partial_1 \gamma(c) \rangle = \langle \rho^* y, c \rangle.$$

Finite-dimensional orthogonality $(\ker \partial_2)^\perp = \text{im } \partial_2^\top$ proves (32). $\square$

Lemma A.2 (Uniform fibre span). *The subspace H_p has dimension at most one.*

Proof. For a coarse point a , let P_a be its line-coordinate pencil. For a coarse direction d , let $R_d(a)$ be the line of direction d through a . Identify the point fibre over a with k^2 , choose a nonzero functional β with kernel d , and give the fibre point b weight $-\beta(b)^{p-1}$. A fine line in direction d has weight sum zero; otherwise β runs through k and $-\sum_{t \in k} t^{p-1} = 1$. Its fine Radon transform is therefore

$$\rho^*(P_a - e_{R_d(a)}) \in \text{Col}_2.$$

These vectors span the total-coordinate-sum-zero hyperplane W_0 . To see this, a vector orthogonal to all of them has equal values on all lines through each point. Intersecting nonparallel lines, and connecting parallel lines by a transversal, forces the vector to be constant; conversely constants are orthogonal to every displayed generator. Thus $Y \cap W_0 \subset H_p^\perp$ by (32). For $y \in Y$, the common parallel sum equals the total coordinate sum because $p+1 = 1$ in k , while P_a has common sum one. Hence $Y \cap W_0$ has codimension one in Y , proving the lemma. $\square$

Theorem A.3 (Saturation and augmentation). *The saturation invariant is*

$$u_p = 0 \quad \text{for odd } p, \quad u_2 = 1.$$

Thus

$$\mathcal{K}_2 \cong \begin{cases} (\mathbb{Z}/p\mathbb{Z})^{p^2(p^2-p+2)/2} \oplus (\mathbb{Z}/p^2\mathbb{Z})^{p^2(p^2-5)/4}, & p \text{ odd}, \\ (\mathbb{Z}/2\mathbb{Z})^6, & p = 2. \end{cases}$$

Proof. By theorem A.2, it remains to decide one point pencil. In the bundle coordinates of theorem 6.1, the kernel of the fine Radon map on bundle d is $\mathcal{J}_d$. The diagonal quotient map $A_k \rightarrow \bigoplus_d A_k/\mathcal{J}_d$ therefore shows that $\rho^*P_0 \in \text{Col}_2$ is equivalent to the existence of an $f \in A_k$ satisfying the simultaneous congruences

$$f \equiv v_d^D \pmod{(z_d, v_d^p)^p} \quad (d \in \mathbb{P}^1(k)), \quad D = p(p-1). \quad (33)$$

Indeed, in a direction bundle the fine offsets reducing through the origin form the order- p subgroup, whose norm is

$$\sum_{j=0}^{p-1} t_d^{pj} = (t_d^p - 1)^{p-1} = v_d^D.$$

Filter A_k by augmentation degree. Its associated graded ring is

$$S = k[X, Y]/(X^{p^2}, Y^{p^2}).$$

For a regular linear pair (ℓ_d, m_d) ,

$$\text{in}(z_d, v_d^p)^p = (\ell_d, m_d^p)^p.$$

A monomial $\ell_d^a m_d^b$ survives precisely when $a + \lfloor b/p \rfloor \leq p-1$. Consequently, in degree D the surviving classes are m_d^D and $\ell_d m_d^{D-1}$; in every degree $D < e \leq p^2 - 1$ only m_d^e survives, and there is no quotient component above degree $p^2 - 1$.

Assume first that p is odd. With $\ell_a = Y - aX$, $m_a = X$ for $a \in k$ and $\ell_\infty = X$, $m_\infty = Y$, use the degree- D form

$$F_D(X, Y) = X^D + Y^D - 2X^{p-1}Y^{D-p+1} + X^{2p-2}Y^{D-2p+2}.$$

On $X = 1$,

$$F_D(1, T) = 1 + T^{D-2p}(T^p - T)^2.$$

Thus its value is one and its first derivative is zero at every finite k -point; the coefficients of Y^D and XY^{D-1} give the same value and first-jet conditions at infinity. It solves the initial degree of every congruence in (33).

After subtracting a lift of F_D , suppose the first residual degree is $e > D$. Each local quotient is then one-dimensional, so its symbol is an arbitrary list of $p+1$ values on

$\mathbb{P}^1(k)$. Evaluation of degree- e binary forms is onto: for each point d , take the product of the p linear forms vanishing at all other rational points, normalize its nonzero value at d , and multiply by a degree- $(e - p)$ form nonvanishing at d . These are a Lagrange basis. Lift the resulting form and subtract again. Each step strictly raises the residual degree, and the local quotients vanish above $p^2 - 1$; the process therefore terminates after finitely many steps and solves all actual, not merely initial, congruences. Hence $\rho^*P_0 \in \text{Col}_2$ and (32) gives $H_p = 0$ for odd p .

For $p = 2$, we spell out the strictness consequence of (18). Give

$$\Phi : A_k \longrightarrow \bigoplus_d A_k / \mathcal{J}_d, \quad f \longmapsto (f \bmod \mathcal{J}_d)_d$$

the augmentation filtrations. Then

$$\ker(\text{gr } \Phi) = \bigcap_d \text{in}(\mathcal{J}_d), \quad \text{gr}(\ker \Phi) = \text{in}\left(\bigcap_d \mathcal{J}_d\right),$$

so (18) identifies these two spaces. If $\Phi(f)$ has filtration degree at least r but f has initial degree $s < r$, its degree- s symbol lies in $\ker(\text{gr } \Phi)$ and is therefore the initial symbol of some $h \in \ker \Phi$. Replacing f by $f - h$ preserves its image and raises its initial degree. Iteration terminates in the finite augmentation filtration and produces a preimage in degree at least r . Hence Φ is strict.

The right side of (33) starts in degree $D = 2$. Strictness shows that any hypothetical preimage may be chosen with quadratic initial form $aX^2 + bXY + cY^2$. The value and first-jet conditions at 0 and ∞ require $a = c = 1$ and $b = 0$. At the third tangent $\ell = Y - X, m = X$, the same conditions require $a + b + c = 1$ and $b = 0$, contradicting $a + b + c = 0$ in $\mathbb{F}_2$. Thus the pencil is not in the fine column code. Hence $H_2 \neq 0$, while theorem A.2 gives $\dim H_2 \leq 1$. Moreover, the total-coordinate-sum-zero subspace $Y \cap W_0$ is exactly the annihilator of the all-one class in $\mathbb{L}_1/2\mathbb{L}_1$. Therefore H_2 is exactly that augmentation line. Therefore $u_2 = 1$. Substitution in (12) gives the stated profiles of $\mathcal{K}_2$. $\square$

AI-use disclosure and author responsibility

Under the author’s direction, OpenAI Codex, using the GPT-5.6 Sol model, generated the central mathematical development, including the all-prime depth-two Smith profile and the transfer-and-carry analysis, and assisted with the exact companion software and manuscript. The author selected the research direction, challenged and checked the derivations and outputs, set the public boundaries, and takes responsibility for the final content. Same-workflow machine reviews are not human peer review or independent expert verification; the repository’s `AI_USE.md` records the detailed provenance.

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